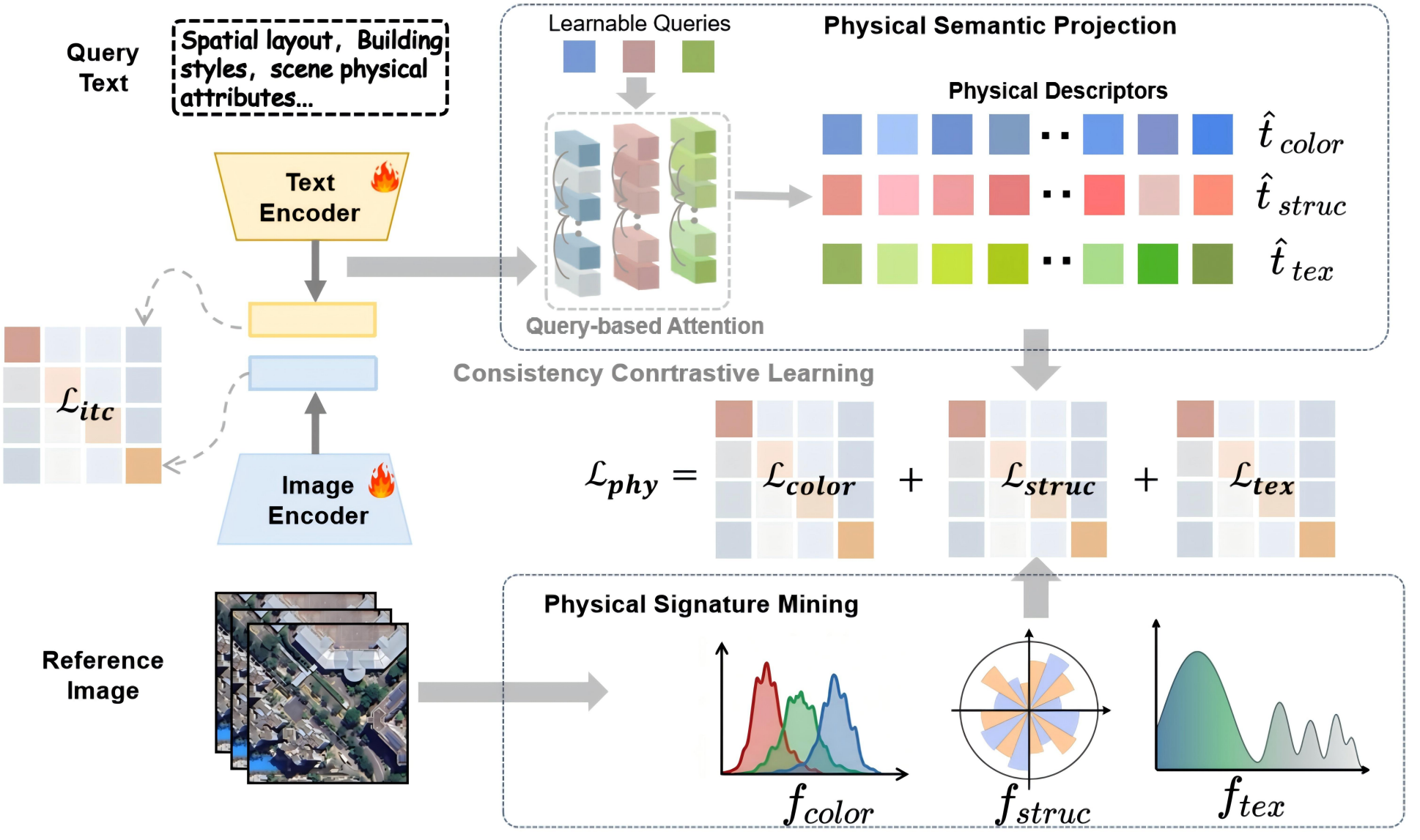
Query Text
Spatial layout, Building styles, scene physical attributes...
Text Encoder
Image Encoder
$\mathcal{L}_{itc}$
Learnable Queries
Query-based Attention
Physical Semantic Projection
Physical Descriptors
$\hat{t}_{color}$
$\hat{t}_{struc}$
$\hat{t}_{tex}$
Consistency Conrtrastive Learning
$\mathcal{L}_{phy} = \mathcal{L}_{color} + \mathcal{L}_{struc} + \mathcal{L}_{tex}$
Reference Image
Physical Signature Mining
f_{color}
f_{struc}
f_{tex}